

Paper 2 - Set 11 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	F	Simultaneous equations and inequalities
2	C	Indices and surds
3	E	Differentiation
4	B	The logic of arguments
5	A	Mathematical proof
6	A	Identifying errors in proofs
7	D	Mathematical proof
8	A	Identifying errors in proofs
9	C	Simultaneous equations and inequalities
10	H	Graphs of functions
11	E	Exponentials and logarithms
12	F	Graphs of functions
13	E	Chains of reasoning
14	E	Trigonometric functions and equations
15	H	Chains of reasoning
16	G	The logic of arguments
17	F	Quadratic functions
18	A	Mathematical proof
19	E	The logic of arguments
20	F	Trigonometric functions and equations

Worked solutions

Question 1 SIM-06 · Simultaneous equations and inequalities · spec MM1.4

Solve $xy = 6$ and $x - y = 1$. What is the sum of all possible values of $x + y$?

Answer: F

$x = y + 1$: $y^2 + y - 6 = 0 \Rightarrow (y + 3)(y - 2) = 0$. Solutions $(3, 2)$ with $x + y = 5$ and $(-2, -3)$ with $x + y = -5$. The sum is 0.

Question 2 IND-15 · Indices and surds · spec MM1.1

Given that $2^x = 5$ and $2^y = 20$, find the value of 2^{2x-y} .

Answer: C

$$2^{2x-y} = \frac{(2^x)^2}{2^y} = \frac{25}{20} = \frac{5}{4}.$$

Question 3 DIF-20 · Differentiation · spec MM6.3

Find the sum of the y -coordinates of the stationary points of the curve $y = 2x^3 - 9x^2 + 12x - 3$.

Answer: E

$y' = 6x^2 - 18x + 12 = 6(x - 1)(x - 2)$: stationary points at $x = 1$ ($y = 2 - 9 + 12 - 3 = 2$) and $x = 2$ ($y = 16 - 36 + 24 - 3 = 1$). Sum 3.

Question 4 LGC-17 · The logic of arguments · spec Arg1

Let x be a real number. Which of the following statements is equivalent to **' $x \geq 2$ if and only if $x^3 \geq 8$ '**?

Answer: B

'If and only if' is the conjunction of the implication and its converse.

Question 5 X2-27 · Mathematical proof · spec Prf1

Consider the claim: **'if n is a positive integer then $n^2 + n + 1$ is odd.'** Which of the following is a valid proof?

Answer: A

A is a direct general argument. B only checks cases; C is irrelevant; D is correct for odd n but says nothing about even n ; E does not reach a contradiction.

Question 6 ERR-10 · Identifying errors in proofs · spec Err2

Claim: if $a > b$ then $\frac{1}{a} < \frac{1}{b}$.

I Suppose $a > b$.

II Dividing both sides by ab : $\frac{1}{b} > \frac{1}{a}$.

III So $\frac{1}{a} < \frac{1}{b}$.

Which of the following is correct?

Answer: A

$a = 1, b = -1$: $a > b$ but $\frac{1}{a} = 1 > -1 = \frac{1}{b}$. Dividing by ab without knowing its sign is invalid (and $b = 0$ would be division by zero).

Question 7 PRF-07 · Mathematical proof · spec Prf1

Prove by cases that $n^3 - n$ is divisible by 3 for every integer n . Which set of cases is both sufficient and necessary to consider?

Answer: D

$n^3 - n = (n - 1)n(n + 1)$ is a product of three consecutive integers; splitting by remainder on division by 3 shows one factor is a multiple of 3 in every case.

Question 8 ERR-18 · Identifying errors in proofs · spec Err2

A student solves $2\cos^2 x = \cos x$ for $0 \leq x < 2\pi$.

I Divide both sides by $\cos x$: $2\cos x = 1$.

II So $\cos x = \frac{1}{2}$.

III So $x = \frac{\pi}{3}$ or $x = \frac{5\pi}{3}$.

How many solutions has the student missed, and where is the first error?

Answer: A

Dividing by $\cos x$ loses $\cos x = 0$, i.e. $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$: two solutions.

Question 9 X2-14 · Simultaneous equations and inequalities · spec MM1.5

x is a real number. P : ' $x^2 < 4$ '. Q : ' $x < 2$ '. Which one of the following is true?

Answer: C

$x^2 < 4$ means $-2 < x < 2$, which implies $x < 2$ (sufficient); but $x = -5$ satisfies Q and not P (not necessary).

Question 10 GRA-05 · Graphs of functions · spec MM8.1

Which of the following statements about the graph of $y = |2x - 4|$ is/are true?

I It meets the x -axis at exactly one point.

II It is symmetric about the line $x = 2$.

III It lies entirely on or above the line $y = 0$.

Answer: H

$|2x - 4| = 0$ only at $x = 2$ (I). $|2x - 4| = 2|x - 2|$ is symmetric about $x = 2$ (II). A modulus is never negative (III).

Question 11 X2-06 · Exponentials and logarithms · spec MM5.2

Which of the following is a counterexample to the claim '**'for all positive x and y ,

$\log_{10}(x + y) \leq \log_{10} x + \log_{10} y$ '**?

Answer: E

$x = y = 1$: $\log 2 > 0 = \log 1 + \log 1$, so the inequality fails. For the others $x + y \leq xy$ and the claim holds (log is increasing).

Question 12 GRA-15 · Graphs of functions · spec MM8.6

The cubic $y = x^3 + ax + b$ has three distinct real roots. Which of the following statements must be true?

- I $a < 0$
- II $b \neq 0$
- III The curve has two turning points.

Answer: F

Three distinct real roots require two turning points (III), which needs $y' = 3x^2 + a$ to have two roots, so $a < 0$ (I). II is false: $x^3 - x$ has roots 0, ± 1 and $b = 0$.

Question 13 RSN-09 · Chains of reasoning · spec Prf5

A rectangle has integer side lengths and its area (in cm^2) is numerically equal to its perimeter (in cm). How many such rectangles are there (counting $a \times b$ and $b \times a$ once)?

Answer: E

$$ab = 2a + 2b \Rightarrow (a - 2)(b - 2) = 4: (a - 2, b - 2) = (1, 4) \text{ or } (2, 2), \text{ giving } 3 \times 6 \text{ and } 4 \times 4.$$

Question 14 TRG-05 · Trigonometric functions and equations · spec MM4.4

Which of the following statements about the graph of $y = \cos x$ is/are true?

- I It is symmetric about the y-axis.
- II It is a translation of the graph of $y = \sin x$ by $\frac{\pi}{2}$ in the negative x -direction.
- III It has period π .

Answer: E

$$\cos(-x) = \cos x \text{ so I is true. } \sin\left(x + \frac{\pi}{2}\right) = \cos x, \text{ so II is true. The period of } \cos \text{ is } 2\pi, \text{ so III is false.}$$

Question 15 RSN-15 · Chains of reasoning · spec Prf5

A sequence of 10 integers is such that the sum of any three consecutive terms is positive and the sum of any five consecutive terms is negative. Which of the following statements must be true?

- I The sum of all 10 terms is negative.
- II The sequence contains at least one negative term.
- III The sequence contains at least one positive term.

Answer: H

Terms 1-5 and 6-10 are two blocks of five, each with negative sum (I). A five-block has negative sum, so some term is negative (II). A three-block has positive sum, so some term is positive (III).

Question 16 LGC-16 · The logic of arguments · spec Arg3

Which of the following statements is/are true about real numbers?

- I For all x and all y , if $x < y$ then $x^2 < y^2$.
- II For all x there exists y such that $x < y$ and $x^2 < y^2$.
- III There exist x and y with $x < y$ and $x^2 > y^2$.

Answer: G

I fails for $x = -2$, $y = 1$. II: take $y = |x| + 1$. III: $x = -2$, $y = 1$ works.

Question 17 QUA-17 · Quadratic functions · spec MM1.3

The quadratic equation $ax^2 + bx + c = 0$ (with $a \neq 0$) has real coefficients. Which of the following statements is/are always true?

- I If $ac < 0$, the equation has two distinct real roots.
- II If $b = 0$, the equation has no real roots.
- III If the equation has a repeated root, then $b^2 = 4ac$.

Answer: F

I: $b^2 - 4ac > 0$ whenever $ac < 0$, so true. II: $x^2 - 4 = 0$ has $b = 0$ and real roots, so false. III: a repeated root means the discriminant is zero, so true.

Question 18 PRF-10 · Mathematical proof · spec Prf1

Consider the claim: **if x and y are real numbers with $x + y > 10$, then $x > 5$ or $y > 5$.** Which of the following proves it?

Answer: A

A is proof by contrapositive/contradiction. B assumes the conclusion, C is an example, D proves something false, E's conclusion ('both exceed 5') is wrong.

Question 19 LGC-20 · The logic of arguments · spec Arg1

Which of the following statements is/are logically equivalent to '**if it is raining, then the ground is wet**'?

- I It is not raining, or the ground is wet.
- II If the ground is not wet, then it is not raining.
- III If the ground is wet, then it is raining.

Answer: E

' $A \Rightarrow B$ ' is equivalent to 'not A , or B ' (I) and to its contrapositive (II). III is the converse, which is not equivalent.

Question 20 TRG-17 · Trigonometric functions and equations · spec MM4.5

Which of the following is/are true for all real x ?

I $\sin^2 x + \cos^2 x = 1$

II $\sin(2x) = 2\sin x$

III $\sin(x + \pi) = -\sin x$

Answer: F

I is the Pythagorean identity. II is false (e.g. $x = \frac{\pi}{2}$: $0 \neq 2$). III: translating $\sin$ by half a period negates it, so III is true.